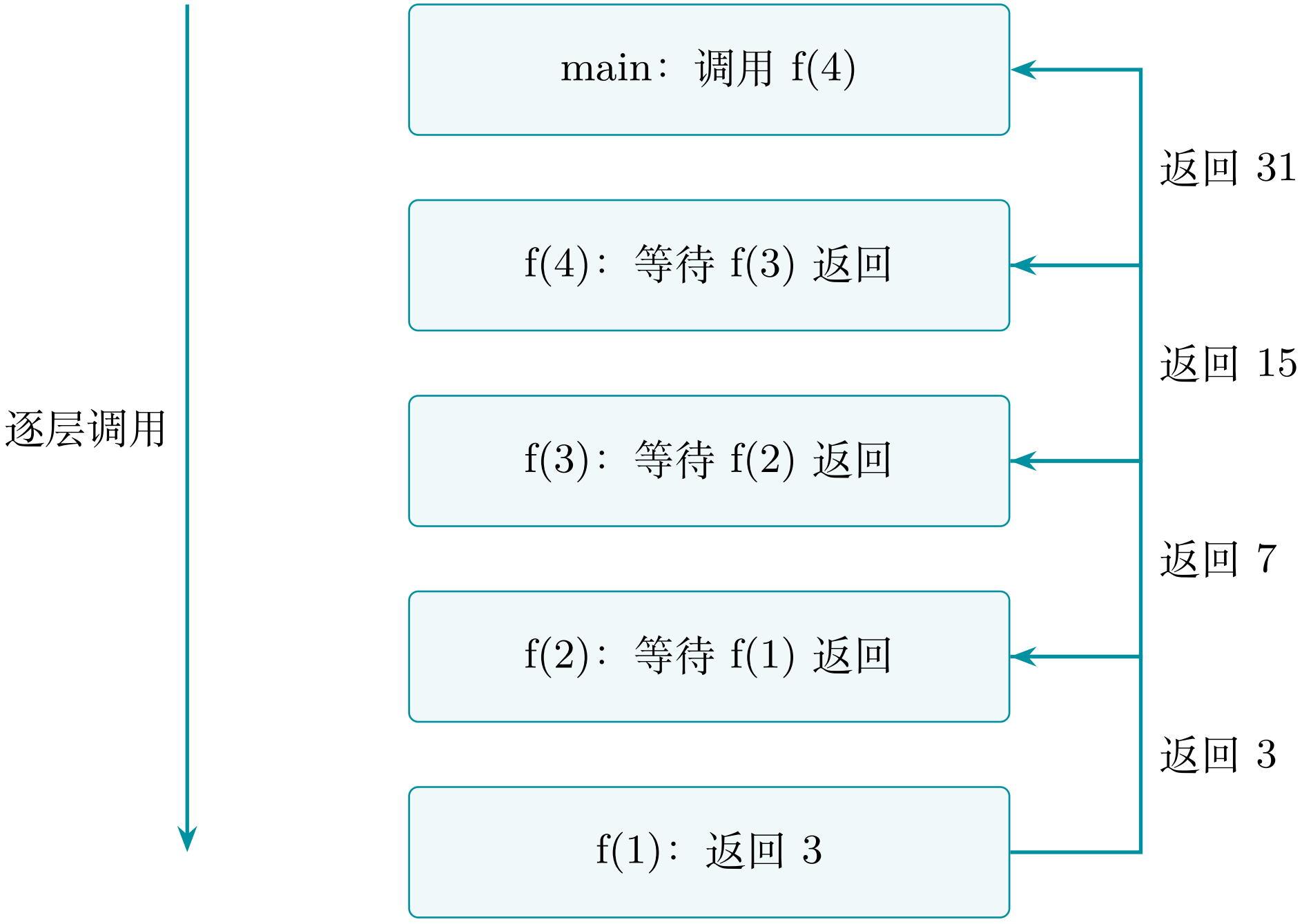

$$f(1)=3;\ \ f(n)=2f(n-1)+1,\ \ n>1$$